

François De Keersmaecker

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PUBLICATIONS

- [1] François De Keersmaecker, Rémi Van Boxem, Cristel Pelsser, and Ramin Sadre. The Forest Behind the Tree: Revealing Hidden Smart Home Communication Patterns. In *2025 IEEE 33rd International Conference on Network Protocols (ICNP)*, pages 1–12, 2025.
- [2] François De Keersmaecker, Ramin Sadre, and Cristel Pelsser. Supervising Smart Home Device Interactions: A Profile-Based Firewall Approach. In *2024 IFIP Networking Conference (IFIP Networking)*, pages 413–422, 2024.
- [3] François De Keersmaecker, Yinan Cao, Gorby Kabasele Ndonga, and Ramin Sadre. A Survey of Public IoT Datasets for Network Security Research. *IEEE Communications Surveys & Tutorials*, 25(3):1808–1840, 2023.

REPOSITORIES

- smart-home-firewall
- firewall-blocklist
- profile-translator-blocklist
- smart-home-testbed
- pcap-fuzzer
- pyyaml-loaders
- lorawan-smart-lln

AWARDS

BEST POSTER AWARD

TMA 2024 PhD School

May 22, 2024 | Dresden, Germany

BEST POSTER

GRASCOMP 2024 Doctoral Day

Sept. 9, 2024 | Louvain-la-Neuve, Belgium

SKILLS

LANGUAGES

🇫🇷 French: native • 🇬🇧 English: C2

🇳🇱 Dutch: B1

PROGRAMMING

Shell • Assembly • C • C++ • Java • Python
Ruby • Scala • JavaScript • Android • Matlab
⌚ Erlang • SystemVerilog

TOOLS

Linux • Windows • Git • SQL • Docker • CI/CD
VirtualBox • Vagrant • GDB • Jinja • Arduino
Ghidra • Zeek • LTspice • LabView

DESKTOP

Word • Excel • PowerPoint

OTHERS

🚗 Driving license: Category B

EXPERIENCE

UCLouvain | TEACHING ASSISTANT

September 2021 - January 2026 | Louvain-la-Neuve, Belgium

- Computer System Security (Graduate)
- Computer Systems (Undergraduate)
- Programming Languages Paradigms (Undergraduate)
- Bases of Computer Science II (Undergraduate)

UCLouvain | DEVELOPER AND COURSE ADMINISTRATOR

Summer 2021 | Louvain-la-Neuve, Belgium

- Updated the syllabus and created new exercises for a computer science course
- Extensively used Docker containers for tool configuration and development
- Extended the features of VSCode for the Oz programming language

UCLouvain | ACADEMIC TUTOR

Sept. 2019 - May 2021 | Louvain-la-Neuve, Belgium

- Supervised practical sessions of bachelor students
- Graded students' assignments
- Had meetings with teachers every week to manage the course

CITY JOB | ANDROID DEVELOPER

March 2019 - July 2019 | Louvain-la-Neuve, Belgium

- Worked with 3 other students on the creation of a start-up
- Developed an Android application to be launched on the Play Store
- Participated in the 11th edition of Mind & Market, to present the start-up to potential investors

EDUCATION

PH.D IN COMPUTER ENGINEERING

UCLouvain | ICTEAM - INGI

Grad. April 2026 | Louvain-la-Neuve, Belgium

F.R.S.-FNRS candidate

Thesis: Profile-Based Security and In-Depth Characterization of Smart Home Network Communications

MS IN COMPUTER ENGINEERING

UCLouvain | LOUVAIN SCHOOL OF ENGINEERING

Grad. June 2021 | Louvain-la-Neuve, Belgium

Major: Networks & Security

Languages: courses fully in English

Summa Cum Laude (Cum. GPA: 87.6 %)

BS IN ENGINEERING

UCLouvain | LOUVAIN SCHOOL OF ENGINEERING

Grad. June 2019 | Louvain-la-Neuve, Belgium

Major: Computer Engineering | Minor: Electrical Engineering

Languages: core courses in French, English language courses

Magna Cum Laude (Cum. GPA: 79.3 %)

SECONDARY SCHOOL

INSTITUT SAINT-JOSEPH

Grad. June 2016 | Ciney, Belgium

Majors: Mathematics & Science

Languages: core courses in French, Dutch in immersion classes, English
Excellence prize (Best student of the promotion)